

XRPL Validator — Daily Engineering Report • 2026-08-31

Compiled retrospectively on 2026-09-23 from the git history. Scope: XRPL work only — the git history of [xrpl-validator](#) and the XRPL Rust SDK, the XRP Ledger itself (amendments) and XRPL research.

Summary

- **21 commits** (xrpl-validator 21)
- 1 test / vector commit

Engineering changes

- **fix(node): drop the stale mirror eagerly — un-deadlock the hydrate budget gate (shadow v6)** (6a853fd , xrpl-validator) The 2026-08-30 overnight receipts exposed a self-deadlock: the first hydration completed (19.8M objects, 220s), a catch-up batch gapped past the mirror, and every re-hydrate for 18 hours was refused by the memory budget gate — because the ~14GB corpse of the orphaned mirror was itself holding the RAM the gate was waiting to see free (108 refusals, MemAvailable pinned at 11GB).
- **feat(node): shadow v7 — self-classifying receipts (pre-audit, noop judge, as_of stamp)** (766214c , xrpl-validator) The finding-38 postmortem: two live byte-diffs (OfferCreate RLUSD line, AMMWithdraw PLX lines) both probed CLEAN with DX_VALCHECK armed, and the codec round-trips the diffed object exactly (examples/roundtrip_entry.rs) — engine and codec exonerated, but the mirror-input evidence was already destroyed by the reconcile that heals each divergence. Three upgrades so the next event arrives self-classified: - ...
- **fix(node): shadow v7.1 — remove the lag guard that livelocked post-hydration catch-up** (f269d43 , xrpl-validator) The gap-fill loop builds a REAL single-ledger overlay per iteration, so the shadow can follow catch-up sequentially. v7's lag<=2 compare guard skipped every backlog ledger the 123s hydration block created, froze at_seq, and the first at-edge compare manufactured the very gap the guard meant to avoid: hydrate -> 3 compares -> gap -> drop -> 600s cooldown -> repeat. The catch-up diverges that motivated the guard were ...
- **fix(node): shadow v7.2 — stretch the FFI catch-up budget when a mirror is live** (861a5cb , xrpl-validator) Second layer of the hydration livelock: skip_shadow_for_catchup fired at lag>25, and hydration's ~123s block leaves ~31 ledgers of backlog — the first six skip the FFI apply, the mirror starves of overlays, position freezes, and the first overlay ledger gaps it. With a hydrated mirror the budget rises to 200 (any hydration wake, with margin); deeper catch-ups gap-drop-rehydrate as designed.
- **fix(ledger): refuse single-asset AMM deposits whose LP tokens adjust to zero (finding 39)** (632e38f , xrpl-validator) First genuine engine bug caught, classified, and prosecuted by the live shadow's own receipts. #106668165 3691EF47: AMMDeposit tfSingleAsset of ONE DROP of XRP into the XRP/SPY pool — rippled refuses with tecAMM_INVALID_TOKENS the moment adjustLPTokensOut rounds the dust to zero (singleDeposit, fixAMMv1_3 arm; deposit()'s epilogue re-checks lpTokensDepositActual <= 0). Our single_adj closure returned None on ...

- **fix(ledger): a zero fill examines an offer, it does not move it (finding 40)** (7f28229 , xrpl-validator) Live shadow catch #106668867 575275AC, self-classified PRE-UNKNOWN (no-meta): a crossing walked past a 2023-era junk offer (max-value self-issued pair) and our settle path rewrote it with identical amounts — a no-op whose only effect was a fresh threading stamp at offset 14 (PreviousTxnLgrSeq). Canonical mainnet still carries the ancient stamp: rippled leaves examined-but-unmoved offers untouched. Zero-fill settles ...
- **feat(scripts): hourly shadow-scout — live-engine divergence digest to telegram** (051a1bc , xrpl-validator) The probe scout cannot see the live shadow's finding class (39 exits early by luck in fixture worlds, 40 needs the fossil offer resident), so the shadow's own receipts become the scout feed: hourly digest of new jsonl entries classified PRE-OK (engine finding) / STALE-PRE-UNKNOWN (mirror input) / ter-mismatch (urgent), plus an advancement check — the 2026-08-31 watcher gap where death and divergence were covered but ...
- **fix(ledger): AccountDelete carries out the nonObligationDeleter cascade instead of refusing on OwnerCount (finding 41)** (7ea26b3 , xrpl-validator) Live shadow catch #106669431 535CFFF6: an account holding one deletable NFTokenOffer (OwnerCount 1) was refused with tecNO_PERMISSION where mainnet deletes it. The preclaim walk already refuses real obligations per entry; the OwnerCount>0 count-gate that stood in for the unmodeled deleter is gone, and do_apply now deletes what the walk passed: plain offers leave their book via delete_maker_offer, NFT offers leave ...
- **feat(node): shadow leak forensics — retry-repair verifier, hydrate integrity audit, valued stale receipts** (3e52c6e , xrpl-validator) The reconcile-leak class (5/day, all created-then-deleted book pages) reads back wrong bytes in the live process while the same bytes roundtrip EXACT in every offline process on both machines; fb==canonical is hash-proven. Only live forensics can name the mechanism: every verifier failure now dumps key/fb/jb/re-encode (an encode ERROR becomes a named verdict), re-asserts the canonical value, and reports retry_fixed ...
- **fix(loop): probe/diff paths gate at 4GB and drop to -j3 under the hydrated mirror** (8c022b8 , xrpl-validator) The live validator's mirror pins MemAvailable ~7.5GB; the flat 8GB gate starved every probe (3x300s wait then exit 71). -j3 peaks ~3GB — the manual bypass from the triage-law memo becomes the built-in fallback. Gate suites and unit runs keep the 8GB gate for their -j6 batteries. First time runner.sh is tracked; deployed copy on the validator host updated in lockstep.
- **feat(node): va-06 publisher-verified UNL — manifest chain, blob signature, sequence gate (S3)** (c258ed6 , xrpl-validator) verify_vl proves a fetched validator list end to end the way rippled's ValidatorList does: optional publisher pin (XRPL_UNL_PINNED_KEY), publisher manifest chain ('MAN\0' preimage minus both signature fields; ed25519 raw / secp256k1 sha512Half digest — the S1/S2 dispatch), blob signature under the manifest's ephemeral key, expiry, and each validator entry's own manifest before its signing key is believed (closes ...
- **fix(ledger): a leaf at depth 64 is legal — stop erroring overwrites of last-nibble sibling keys** (2670c0c , xrpl-validator) Two distinct keys sharing their first 63 nibbles (adjacent book-page qualities, constantly) legally place both leaves at depth 64. The old top-of-insert guard let the pair be BUILT, then errored every LATER overwrite of either key — and every 'let _ = insert(...)' caller silently lost the write while lookup/delete (unguarded) kept serving the stale leaf. Live signature (2026-08-31): a book page froze at its previous ...
- **feat(node): count every mirror map-op failure (map_op_err) instead of swallowing it** (53f6169 , xrpl-validator) The depth-64 freeze hid for hours behind 'let _ = insert(...)'. Every state_map insert/delete in the shadow now feeds note_map_op; the counter is on /api/engine and must read 0 forever — any tick is a structural regression.
- **fix(node): canon_for_encode learns Subject and the other post-2024 AccountID/u64 SFields** (965a7a9 , xrpl-validator) The hydrate integrity audit's first run named 386 of 19.8M objects unencodable — every Credential in

the mirror: 'Subject' was missing from canon's ACCTS table, so its 40-hex value reached the encoder and was parsed as base58 (InvalidBase58/InvalidAddress). Census vs definitions.json adds all seven missing AccountID names (Subject, Delegate, Counterparty, Borrower, OtherChainDestination, EmitCallback, HookAccount) ...

- **fix(node): ter-mismatch stale audit compares amount objects by value+currency, not strict JSON** (75d553e , xrpl-validator) Strict equality branded byte-identical balances STALE because the mirror dialect spells issuers as hex where the meta has r-addresses — four path ter-mms in the post-fix soak eve were mislabeled input-caused when the displayed values matched exactly. Same comparison rule pre_stale_audit has always used.
- **fix(ledger): AMMCreate builds the complete canonical object set (finding 46)** (2316d8b , xrpl-validator) Five sub-defects, all named by mainnet #106674486 (tx 8A6C485D) and the new byte-exact vector gate: (1) initialize_fee_auction_vote — the faithful port that already served the deposit-revival path — was never called at creation, so AuctionSlot/VoteSlots stayed empty placeholders (AMMCreate.cpp:260); (2) the AMM object lacked its explicit zero Flags; (3) it was never dirLinked into the AMM account's own owner ...
- **fix(ledger): EscrowCreate stamps the creator's Sequence on the escrow object (finding 45)** (b8bfeea , xrpl-validator) rippled keys the escrow AND writes getSeqValue() into it (Escrow.cpp:531+544) — the tx sequence, or the ticket sequence for a ticketed create; ours wrote neither the field nor the ticket arm of the keylet (offer.rs:5613 idiom now shared). Live shadow #106674441: ours 114B vs canonical 119B, diff @8 = exactly the missing 0x24 UInt32. Gate: escrow_create_vector replays the real mainnet transaction against its real ...
- **fix(ledger): the synthetic AMM offer's partial-fill rate is getRate's half-up, not a ceiling (finding 47)** (a80e516 , xrpl-validator) Mainnet #106674439 tx 81F71E9D (tfPartial XJOY → CBIRD → XRP): thirteen flow iterations aligned digit-identical against rippled's own trace except the last — the only PARTIAL take of a synthesized AMM offer (full takes carry ofr_in verbatim and never round). rippled prices it through AMMOffer::limitOut → Quality::ceil_out_strict → mulRoundStrict(limit, quality.rate(), Up), and quality.rate() = getRate = ...
- **fix(ledger): NFT IOU sales pay every leg as a buyer-side accountSend, grossed by the token issuer's TransferRate (finding 48)** (dcf5d02 , xrpl-validator) Mainnet #106677548 (NFTTokenAcceptOffer, 950,000 HADA direct sale, 2.5% NFT royalty): the buyer ended 475 HADA rich — exactly 950,000 x 0.0005, the HADA issuer's TransferRate (1000500000). rippled pays broker cut, NFT royalty and seller as separate accountSends FROM THE BUYER (NFTTokenAcceptOffer.cpp:491-503, 570-596), and accountSend grosses the sender's debit by the leg issuer's rate; our pay_iou_with_transfer_fee ...
- **fix(ledger): AMMVote applyVote ported line-for-line — five modeled rules corrected (finding 49)** (603c184 , xrpl-validator) Mainnet #106678645 (XPM/XRP AMMVote): ours collapsed an 8-slot vote slate to the voter alone and wrote TradingFee 1000 where mainnet holds 761. The vendored AMMVote.cpp arbitrates five rules the old code modeled: (1) only a holder whose LP BALANCE is zero drops — dust whose weight ROUNDS to zero is retained at VoteWeight 0; (2) the voter's entry updates IN PLACE, keeping slot order; (3) TradingFee averages fee x ...

Tests and campaigns

- **test(ledger): fuzz ordered prefix enumeration against the oracle under churn** (1081e08 , xrpl-validator)

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **Fan-Out Batch Claims for Payment Channels** (gaming) Add a `tfMultiDest` flag to `PaymentChannelCreate` that creates a `PayChannel` with no fixed `sfDestination`, authorized instead by the channel's existing `sfP`
- **Credential-Bound Trust-Line Authorization Expiry** (compliance) Add an optional `sfAuthExpiration` (uint32 ledger close time) and `sfAuthCredentialID` (Hash256, referencing an existing `Credential` keylet) to `RippleState`
- **Bonded Keeper-Executed Scheduled Transactions** (hooks) Add a new `ltSCHEDULED_TX` object (owner-reserved, keylet::scheduled(account, sequence)) created by `ttSCHEDULE_CREATE`, storing a fully pre-signed inner

Appendix — commits

Time	Repo	Commit	Subject
07:28	xrpl-validator	6a853fd	fix(node): drop the stale mirror eagerly — un-deadlock the hydrate budget gate (shadow v6)
09:08	xrpl-validator	766214c	feat(node): shadow v7 — self-classifying receipts (pre-audit, noop judge, as_of stamp)
09:32	xrpl-validator	f269d43	fix(node): shadow v7.1 — remove the lag guard that livelocked post-hydration catch-up
09:44	xrpl-validator	861a5cb	fix(node): shadow v7.2 — stretch the FFI catch-up budget when a mirror is live
10:30	xrpl-validator	632e38f	fix(ledger): refuse single-asset AMM deposits whose LP tokens adjust to zero (finding 39)
10:59	xrpl-validator	7f28229	fix(ledger): a zero fill examines an offer, it does not move it (finding 40)
11:08	xrpl-validator	051a1bc	feat(scripts): hourly shadow-scout — live-engine divergence digest to telegram
11:59	xrpl-validator	7ea26b3	fix(ledger): <code>AccountDelete</code> carries out the <code>nonObligationDeleter</code> cascade instead of refusing on <code>OwnerCount</code> (finding 41)
13:23	xrpl-validator	3e52c6e	feat(node): shadow leak forensics — retry-repair verifier, hydrate integrity audit, valued stale receipts
13:25	xrpl-validator	8c022b8	fix(loop): probe/diff paths gate at 4GB and drop to -j3 under the hydrated mirror
13:34	xrpl-validator	c258ed6	feat(node): va-06 publisher-verified UNL — manifest chain, blob signature, sequence gate (S3)
14:00	xrpl-validator	2670c0c	fix(ledger): a leaf at depth 64 is legal — stop erroring overwrites of last-nibble sibling keys
14:00	xrpl-validator	53f6169	feat(node): count every mirror map-op failure (<code>map_op_err</code>) instead of swallowing it
14:00	xrpl-validator	965a7a9	fix(node): <code>canon_for_encode</code> learns <code>Subject</code> and the other post-2024 <code>AccountID/u64</code> SFields
16:30	xrpl-validator	75d553e	fix(node): ter-mismatch stale audit compares amount objects by value+currency, not strict JSON
16:36	xrpl-validator	1081e08	test(ledger): fuzz ordered prefix enumeration against the oracle under churn

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18:12	xrpl-validator	2316d8b	fix(ledger): AMMCreate builds the complete canonical object set (finding 46)
18:12	xrpl-validator	b8bfeea	fix(ledger): EscrowCreate stamps the creator's Sequence on the escrow object (finding 45)
19:24	xrpl-validator	a80e516	fix(ledger): the synthetic AMM offer's partial-fill rate is getRate's half-up, not a ceiling (finding 47)
21:30	xrpl-validator	dcf5d02	fix(ledger): NFT IOU sales pay every leg as a buyer-side accountSend, grossed by the token issuer's TransferRate (finding 48)
22:16	xrpl-validator	603c184	fix(ledger): AMMVote applyVote ported line-for-line — five modeled rules corrected (finding 49)